

Part 1

Set Theory

Definition	Statement
Complement	$P(A^c) = 1 - P(A)$
Monotonicity	If $A \subset B$, then $P(A) \leq P(B)$
Inclusion–Exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
De Morgan’s Law	$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B)$
Countable Additivity	For disjoint A_i , $P(\bigcup_i A_i) = \sum_i P(A_i)$
Difference of events	$P(A \setminus B) = P(A \cap B^c) = P(A) - P(A \cap B)$
Symmetric difference	The event exactly one of A and B occurs: $A \Delta B = (A \setminus B) \cup (B \setminus A)$, and $P(A \Delta B) = P(A) + P(B) - 2P(A \cap B)$.

Problem-Solving Considerations

- **Formalize Ω and $\mathcal{F}$:** Identify what represents a single outcome.
- **Translate to set operations:** Convert English descriptions to $A, B, \cup, \cap$, and c .
- **Double inclusion:** To prove $A = B$, show both $A \subseteq B$ and $B \subseteq A$. Equivalently, show $\omega \in A \Rightarrow \omega \in B$ and $\omega \in B \Rightarrow \omega \in A$.
- **Limsup/Liminf events:** $\omega \in \limsup_n A_n \Leftrightarrow \forall n \exists m \geq n : \omega \in A_m$ (infinitely often), while $\omega \in \liminf_n A_n \Leftrightarrow \exists n \forall m \geq n : \omega \in A_m$ (eventually always).
- **Identify disjointness:** Check if $A \cap B = \emptyset$ before summing.
- **Decompose when needed:** If not disjoint, use inclusion–exclusion or subtract intersections.
- **Induced measures:** If a random variable is involved, use $F_X(x) = P(X \leq x)$ to map sets from $\mathcal{F}$ to the Borel sets on $\mathbb{R}$.
- **Indicator functions:** Use indicators to rewrite set statements as expectations or sums. For example, $P(A) = E[\mathbf{1}_A]$ and $P(\cup_i A_i) = E[\sum_i \mathbf{1}_{A_i}]$.
- **Basic extension:** For any event A and sub- σ -algebra $\mathcal{F}$, $E[\mathbf{1}_A | \mathcal{F}] = P(A | \mathcal{F})$.
- **Union as a conditional sum:** If $\{A_i\}$ are pairwise disjoint, then

$$\mathbf{1}_{\cup_i A_i} = \sum_i \mathbf{1}_{A_i} \implies P\left(\bigcup_i A_i \middle| \mathcal{F}\right) = \sum_i P(A_i | \mathcal{F}).$$

More generally, ” $=$ ” becomes ” $\leq$ ” for non-disjoint events. For two events,

$$P(A \cup B | \mathcal{F}) = P(A | \mathcal{F}) + P(B | \mathcal{F}) - P(A \cap B | \mathcal{F}).$$

- **Law of Total Probability:** If $\{B_j\}$ is a partition with $P(B_j) > 0$,

$$P(A | \mathcal{F}) = \sum_j P(A | B_j) P(B_j | \mathcal{F}).$$

Similar for expectations (law of total expectation): for any integrable r.v. X ,

$$E[X | \mathcal{F}] = \sum_j E[X | B_j] P(B_j | \mathcal{F}).$$

For sub- σ -algebras $\mathcal{G} \subset \mathcal{F}$,

$$E[E[X | \mathcal{F}] | \mathcal{G}] = E[X | \mathcal{G}], \quad E[E[X | \mathcal{G}]] = E[X].$$

Using indicator functions one can expand probabilities or expectations via the partition: if A is an event,

$$\begin{aligned} P(A) &= E[\mathbf{1}_A] = \sum_j E[\mathbf{1}_A \mathbf{1}_{B_j}] \\ &= \sum_j P(A \cap B_j) = \sum_j P(A | B_j) P(B_j), \end{aligned}$$

and conditionally

$$E[\mathbf{1}_A | \mathcal{F}] = \sum_j E[\mathbf{1}_A \mathbf{1}_{B_j} | \mathcal{F}] = \sum_j P(A | B_j) P(B_j | \mathcal{F}).$$

Important Definitions

- **Probability space:** A triple $(\Omega, \mathcal{F}, P)$ where Ω is the sample space, $\mathcal{F}$ a σ -algebra of subsets of Ω , and P a probability measure on $\mathcal{F}$.
- **Event:** An element of $\mathcal{F}$ (a subset of Ω whose probability is defined).
- **Random variable:** A measurable function $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$; i.e. for every Borel set B , $X^{-1}(B) \in \mathcal{F}$.
- **Measurability:** X is measurable w.r.t. a sub- σ -algebra $\mathcal{G} \subset \mathcal{F}$ if $X^{-1}(B) \in \mathcal{G}$ for all Borel B .
- **Borel σ -algebra $\mathcal{B}(\mathbb{R})$:** The smallest σ -algebra containing all open intervals in $\mathbb{R}$.
- **Atom:** An event $A \in \mathcal{F}$ with $P(A) > 0$ such that for any $B \subset A$, $B \in \mathcal{F}$ implies $P(B) = 0$ or $P(B) = P(A)$.
- **Distribution / Pushforward measure:** The law of X is the measure $\mu_X(B) = P(X \in B) = P(X^{-1}(B))$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.
- **Cumulative distribution function (CDF):** $F_X(x) = P(X \leq x)$, right-continuous and non-decreasing with limits 0 and 1.
- **Independence (events):** Events A and B are independent if $P(A \cap B) = P(A)P(B)$. A collection is independent if every finite subcollection is independent.
- **Pairwise vs. mutual independence:** Pairwise independence means every pair is independent. Mutual independence means every finite subcollection satisfies e.g. $P(X, Y, Z) = P(X=x) \cdot P(Y=y) \cdot P(Z=z)$. Mutual independence $\Rightarrow$ pairwise independence, but not conversely.

- **Independence (r.v.’s):** Random variables X and Y are independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all Borel sets A, B .
- **Indicator function:** $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$, else 0. Useful to express probabilities and expectations: $P(A) = E[\mathbf{1}_A]$.
- **Conditional probability:** For $P(B) > 0$, $P(A | B) = P(A \cap B) / P(B)$; extends via regular conditional probabilities when conditioning on σ -algebras or r.v.’s.
- **Filtration:** An increasing family of σ -algebras $\{\mathcal{F}_t\}_{t \in T}$ representing information growth over time.

Common discrete distributions

- **Bernoulli(p):** Single trial with outcomes 1 (success) w.p. p and 0 (failure) w.p. $1 - p$. $P(X = 1) = p$, $P(X = 0) = 1 - p$. Mean $E[X] = p$, variance $\text{Var}(X) = p(1 - p)$.
- **Binomial(n, p):** Sum of n independent Bernoulli(p) trials. $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, \dots, n$. Mean $E[X] = np$, variance $\text{Var}(X) = np(1 - p)$.
- **Poisson(λ):** Counts of rare events in a fixed interval. $P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$, $k = 0, 1, \dots$. Mean and variance both equal λ .
- **Geometric(p):** Number of trials until the first success (support depends on convention). Using the support $k = 1, 2, \dots$: $P(X = k) = p(1 - p)^{k-1}$. Mean $E[X] = 1/p$, variance $\text{Var}(X) = (1 - p)/p^2$.

Common continuous distributions

- **Exponential(λ):** Standard for modeling interevent times (sojourn times) in Markov chains. $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$. Mean $E[X] = 1/\lambda$, variance $\text{Var}(X) = 1/\lambda^2$. Possesses the unique memoryless property.
- **Gamma(α, β):** A generalization of the exponential distribution (sum of α exponentials). Shape parameter $\alpha > 0$, rate parameter $\beta > 0$. Mean $E[X] = \alpha/\beta$, variance $\text{Var}(X) = \alpha/\beta^2$.
- **Normal(μ, σ^2):** The limit of many independent random variables (CLT). $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$. Mean $E[X] = \mu$, variance $\text{Var}(X) = \sigma^2$.

Expectation, variance and conditional expectation

- **Expectation:** For a discrete r.v. X with p.m.f. p_X , $E[X] = \sum_x x p_X(x)$ and $E[g(X)] = \sum_x g(x) p_X(x)$

For a continuous r.v. with p.d.f. f_X ,

$$E[X] = \int_{\mathbb{R}} x f_X(x) dx \quad \text{and} \quad E[g(X)] = \int_{\mathbb{R}} g(x) f_X(x) dx$$

- **Variance:** $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$. For constants a, b , $E[aX + b] = aE[X] + b$ and $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- **Indicator functions:** For an event A , $P(A) = E[\mathbf{1}_A]$. Also $E[X \mathbf{1}_A]$ is the contribution of X on the event A . If $P(A) > 0$, then $E[X | A] = \frac{E[X \mathbf{1}_A]}{P(A)}$. In the discrete case, $E[X | A] = \frac{\sum_{x \in A} x P(X = x)}{P(A)}$.
- **Conditional expectation on a partition:** If $A_1, \dots, A_m$ form a partition and $\sigma(Y)$ has atoms A_i , then $E[X | Y] = \sum_{i=1}^m E[X | A_i] \mathbf{1}_{A_i}$. Equivalently, $E[X | Y](\omega) = \Psi(Y(\omega))$, where $\Psi(y_i) = E[X | Y = y_i] = E[X | A_i]$.
- **Law of total expectation:** $E(X) = E(E(X | Y))$.
- **Law of total variance:** $\text{Var}(X) = E(\text{Var}(X | Y)) + \text{Var}(E(X | Y))$. Hence, $\text{Var}(X) \geq \text{Var}(E(X | Y))$.

Useful distribution tricks

- **CDF method:** For a random variable Y , the c.d.f. is $F_Y(x) = P(Y \leq x)$. Often it is easier to compute the survival probability $F_Y(x) = 1 - P(Y > x)$. If F_Y is differentiable, then $f_Y(x) = \frac{d}{dx}(F_Y(x))$. Always check the support, e.g. if $Y \geq 0$, then $F_Y(x) = 0$ and $f_Y(x) = 0$ for $x < 0$.
- **Minimum trick:** For independent $X_1, \dots, X_n$, $P(\min_i X_i > x) = P(X_1 > x, \dots, X_n > x) = \prod_i P(X_i > x)$. In particular, if $X_i \sim \text{Exp}(\lambda_i)$ are independent, then $\min_i X_i \sim \text{Exp}(\sum_i \lambda_i)$.
- **Convolution:** If $Z = X_1 + X_2$ and X_1, X_2 are independent continuous r.v.’s, then $f_Z(z) = (f_{X_1} * f_{X_2})(z) = \int_{-\infty}^{\infty} f_{X_1}(x) f_{X_2}(z - x) dx$. If $X_1, X_2 \geq 0$ and $f_Z(z) = 0$ for $z < 0$, then usually $f_Z(z) = \int_0^z f_{X_1}(x) f_{X_2}(z - x) dx$.
- **Independent products:** If X_1 and X_2 are independent and integrable, then $E[X_1 X_2] = E[X_1] \cdot E[X_2]$.

Characteristic Function

- The characteristic function of a random variable X is $\phi_X(t) = \mathbb{E}(e^{itX})$.
- For discrete X : $\phi_X(t) = \sum_{x \in R(X)} e^{itx} \mathbb{P}(X = x)$.
- For continuous X : $\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx$.
- Always: $\phi_X(0) = \mathbb{E}(e^{i0X}) = \mathbb{E}(1) = 1$.

Part 2

Stochastic Processes & Markov Property

Core Definitions

- **Stochastic process:** a family $(X_t)_{t \in T}$ of random variables, with T discrete or continuous.
- **Filtration:** an increasing family $(\mathcal{F}_t)_{t \in T}$ with $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$, representing information up to time t .
- **Adapted:** X_t is $\mathcal{F}_t$ -measurable for every t .
- **Natural filtration:** $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$, the smallest fil-

- tration making (X_t) adapted.
- **Markov property:** For discrete time, $P(X_{n+1} \in C | \mathcal{F}_n) = P(X_{n+1} \in C | X_n)$ a.s.

so the present state is a sufficient statistic for the future.

- **Transition Matrix:** For a Markov chain with states S_1, S_2 , the transition matrix is

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix},$$

- where $p_{ij} = \mathbb{P}(X_{n+1} = S_j | X_n = S_i)$. Rows correspond to the current state, columns to the next state, e.g. p_{12} Probability $1 \rightarrow 2$. A transition matrix is stochastic: $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$ for every row i .
- **k-Step Transition Probabilities:** For a Markov chain with transition matrix P , the k -step transition matrix is $P(k) = P^k$. Its entries are:

$$p_{ij}(k) = \mathbb{P}(X_{n+k} = S_j | X_n = S_i) = (P^k)_{ij}.$$

- Thus, the probability of going from state S_i to state S_j in k steps is the (i, j) -entry of P^k .
- **Time-homogeneous chain:** $P(X_{n+1} = j | X_n = i) = p_{ij}$ does not depend on n ; collect these in the transition matrix $P = (p_{ij})$.
- **Chapman–Kolmogorov:** $P^{m+n} = P^m P^n$, and more generally $P(X_{m+n} = j | X_m = i) = (P^n)_{ij}$.
- **Strong Markov property:** at a stopping time τ , the post- τ process depends on the past only through X_τ .
- **CTMC:** a continuous-time process with generator $Q = (q_{ij})$, where $q_{ij} \geq 0$ for $i \neq j$ and $q_{ii} = -\sum_{j \neq i} q_{ij}$.

Stopping Times

- **Stopping time:** τ is a stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ for all t .
- **Discrete-time form:** equivalently, $\{\tau = n\} \in \mathcal{F}_n$ and $\{\tau \leq n\} \in \mathcal{F}_n$ for every n .
- **Stopping-time σ -algebra:** $\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \ \forall t\}$

- and in discrete time this is often checked via $A \cap \{\tau = n\} \in \mathcal{F}_n$.
- **Hitting time:** $\tau_A = \inf\{t \geq 0 : X_t \in A\}$ is a standard example.
- **Stopped process:** $X_{t \wedge \tau}$ is the process frozen at time τ .

Measurability at a Stopping Time

- To show X_τ is $\mathcal{F}_\tau$ -measurable, write for any Borel set B , $\{X_\tau \in B\} = \bigcup_n (\{X_n \in B\} \cap \{\tau = n\})$.
- Since (X_n) is adapted, $\{X_n \in B\} \in \mathcal{F}_n$, and since τ is a stopping time, $\{\tau = n\} \in \mathcal{F}_n$; hence each term lies in $\mathcal{F}_n$.
- Therefore $\{X_\tau \in B\} \in \mathcal{F}_\tau$.

Constructing a Markov State

- If a component X_n depends on a hidden variable Y_n , then (X_n) alone may fail to be Markov.
- Enlarge the state to $Z_n = (X_n, Y_n)$ on the state space $S_X \times S_Y$.
- If the updates factor as $P(Z_{n+1} = (x', y') | Z_n = (x, y)) = P(X_{n+1} = x' | X_n = x, Y_n = y) P(Y_{n+1} = y' | Y_n = y)$,

- then (Z_n) is Markov.
- Stationary distribution: $\pi P = \pi$ for discrete time, or $\pi Q = 0$ for CTMCs.

Part 3

Martingales and Optional Stopping

Notation Formalism

- **Discrete-time martingale:** A process $(M_n)_{n \geq 0}$ is a martingale w.r.t. $(\mathcal{F}_n)$ if for all n : (i) M_n is $\mathcal{F}_n$ -measurable/adapted, (ii) $E[|M_n|] < \infty$, (iii) $E[M_{n+1} | \mathcal{F}_n] = M_n$ a.s.
- **Continuous-time martingale:** A process $(M_t)_{t \geq 0}$ is a martingale w.r.t. $(\mathcal{F}_t)$ if for all $t \geq 0$: (i) M_t is $\mathcal{F}_t$ -measurable/adapted, (ii) $E[|M_t|] < \infty$, (iii) $E[M_t | \mathcal{F}_s] = M_s$ a.s. for all $0 \leq s < t$.
- **Submartingale / supermartingale:** replace equality by $\geq$ or $\leq$ in the conditional expectation relation.

Martingales and conditional expectation

- **Markov vs. martingale:** Markov $\nRightarrow$ Martingale and Martingale $\nRightarrow$ Markov. Markov property: for $s \leq t$, $\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s]$ for bounded f . Martingale property: $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$.
- **(i) Checking measurability/adaptedness:** If M_n is a function of $X_0, \dots, X_n$, then M_n is $\mathcal{F}_n^X$ -measurable, where $\mathcal{F}_n^X = \sigma(X_0, \dots, X_n)$.
- **(ii) Checking integrability:** Finite state space (M_n) takes only finitely many values) or bounded M_n implies integrability. Finite second moment implies integrability by Cauchy–Schwarz: $E[|X|] \leq \sqrt{E[X^2]}$. For sums/differences, use triangle inequality: $E[|X \pm Y|] \leq E[|X|] + E[|Y|]$.
- **(iii) Conditional expectation rules / known vs. future terms:** Let $\mathcal{F}_n^X = \sigma(X_0, \dots, X_n)$. This means that at time n , the variables $X_0, \dots, X_n$ are known, but $X_{n+1}, X_{n+2}, \dots$ are not known.
 - **Deterministic terms:** If c is deterministic, then $E[c | \mathcal{F}_n^X] = c$ and $E[cY | \mathcal{F}_n^X] = cE[Y | \mathcal{F}_n^X]$. In continuous time, $s, t, t-s$ are deterministic, e.g. $E[(t-s)Y_t | \mathcal{F}_s] = (t-s)E[Y_t | \mathcal{F}_s]$.
 - **Linearity:** $E[aY + bZ | \mathcal{F}_n^X] = aE[Y | \mathcal{F}_n^X] + bE[Z | \mathcal{F}_n^X]$.

- **Known terms / taking out what is known:** If Y depends only on $X_0, \dots, X_n$, then Y is $\mathcal{F}_n^X$ -measurable, hence
$$E[Y \mid \mathcal{F}_n^X] = Y$$

Moreover, for any integrable Z , $E[Y Z \mid \mathcal{F}_n^X] = Y E[Z \mid \mathcal{F}_n^X]$. Examples of known terms: X_n , any function $g(X_0, \dots, X_n)$.
- **Future independent terms:** If X_{n+1} is independent of $\mathcal{F}_n^X$, then
$$E[g(X_{n+1}) \mid \mathcal{F}_n^X] = E[g(X_{n+1})]$$

- **Markov shortcut:** If X is Markov, then for $s \leq t$, $E[f(X_t) \mid \mathcal{F}_s] = E[f(X_t) \mid X_s]$. In discrete time: $E[f(X_{n+1}) \mid \mathcal{F}_n^X] = E[f(X_{n+1}) \mid X_n]$.

Problem-Solving Considerations

- Check martingale conditions in order: adaptedness/measurability, integrability, conditional expectation.
- For additive processes, isolate increments and use independence or independent increments.
- For multiplicative processes, factor out the $\mathcal{F}_s$ -measurable part before evaluating the conditional expectation.
- In Markov settings, replace $\mathcal{F}_n$ by the current state X_n when the process is Markovian.
- For hitting-time questions, identify a martingale whose stopped value equals the boundary value you need.
- For expected hitting times, use a second martingale with an explicit time term, such as $S_n^2 - n$ or an It'-motivated transform.
- Always state the stopping-time assumptions needed for OST: bounded τ , integrable stopping time, or uniform integrability of the martingale.

Important Definitions

- **Martingale:** a fair game process; conditional expectation of the future equals the present.
- **Local martingale:** a process that becomes a martingale after stopping at an increasing sequence of bounded stopping times.
- **Uniformly integrable martingale:** a martingale (M_t) with $\{M_t\}$ uniformly integrable; this is a standard hypothesis for OST.
- **Doob decomposition:** a submartingale can be written as a martingale plus an increasing predictable process.
- **Exponential martingale:** if W_t is Brownian motion, then $\exp\left(\sigma W_t - \frac{1}{2}\sigma^2 t\right)$ is a martingale.

Examples of Martingales

- **Simple symmetric random walk:** if $S_n = \sum_{k=1}^n X_k$ with $P(X_k = \pm 1) = 1/2$, then (S_n) is a martingale.
- **Compensated Poisson process:** if (N_t) is Poisson with rate λ , then $(N_t - \lambda t)$ is a martingale.
- **Brownian motion:** (W_t) is a martingale.
- **Brownian square:** $W_t^2 - t$ is a martingale.
- **Cubic Brownian transform:** $W_t^3 - 3tW_t$ is a martingale.
- **Exponential Brownian martingale:** $\exp\left(\sigma W_t - \frac{1}{2}\sigma^2 t\right)$ is a martingale.
- **Stopped martingale:** if (M_t) is a martingale and τ is a stopping time, then $(M_{t \wedge \tau})$ is a martingale.

Optional Stopping Templates

Optional Stopping Theorem (OST): If (M_n) is a martingale and τ is a stopping time satisfying the technical conditions, then

$$E[M_\tau] = E[M_0]$$

Equivalently, stopping a martingale at a valid stopping time preserves its expectation.

When using OST, plug in the possible values of S_τ . If some terms become identical constants, substitute them and treat them as deterministic and take them outside the expectation.

- **Optional sampling:** the discrete-time version of OST for stopping times $\sigma \leq \tau$.
- For symmetric random walk hitting b before $-a$, take $\tau = \inf\{n \geq 1 : S_n \in \{-a, b\}\}$ and apply OST to $M_n = S_n$: $E[S_\tau] = b P(S_\tau = b) - a P(S_\tau = -a)$.
- For expected hitting times, use $M_n = S_n^2 - n$ and apply OST to obtain a linear equation for $E[\tau]$.
- For Brownian motion, use W_t , $W_t^2 - t$, or an exponential martingale depending on the boundary condition.
- For Poisson processes, use $N_t - \lambda t$ when stopping at a jump-related event.

Poisson processes and Brownian motion

- **Poisson process:** $N_t \sim \text{Poisson}(\lambda t)$, so $E[N_t] = \lambda t$, $\text{Var}(N_t) = \lambda t$, and $P(N_t = k) = e^{-\lambda t} \lambda^k t^k / k!$.
- **Poisson increments:** For $0 \leq s < t$, $N_t = N_s + (N_t - N_s)$, where $N_t - N_s \sim \text{Poisson}(\lambda(t - s))$ and is independent of $\mathcal{F}_s$. Hence independent increments property: $E[N_t - N_s \mid \mathcal{F}_s] = E[N_t - N_s] = \lambda(t - s)$

- **Standard Brownian motion:** A process $(W_t)_{t \geq 0}$ is a standard Brownian motion if $W_0 = 0$, it has continuous paths, independent increments, and $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$.
- **Brownian increments:** Write $W_t = W_s + (W_t - W_s)$, where W_s is $\mathcal{F}_s$ -measurable and $W_t - W_s$ is independent of $\mathcal{F}_s$. Hence $E[W_t - W_s \mid \mathcal{F}_s] = 0$, $E[(W_t - W_s)^2 \mid \mathcal{F}_s] = \text{Var}(W_t - W_s) = t - s$.
- **Useful Brownian moments:** Since $W_t \sim \mathcal{N}(0, t)$, $E[W_t] = 0$, $\text{Var}(W_t) = t$, and $E[W_t^2] = \text{Var}(W_t) + (E[W_t])^2 = t$.
- **Brownian covariance:** For $0 \leq s \leq t$, $W_t = W_s + (W_t - W_s)$, so $E[W_s W_t] = E[W_s^2] + E[W_s(W_t - W_s)] = s + 0 = s$. Therefore $\text{Cov}(W_s, W_t) = E[W_s W_t] - E[W_s]E[W_t] = s$. In general, $\text{Cov}(W_s, W_t) = \min(s, t)$.

Part 4

Stochastic Differential Equations & Itô's Lemma

Notation Formalism

- **Itô SDE (scalar):** $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, with initial X_0 .
- **Itô process / diffusion:** process expressible as a drift plus Itô integral.
- **Strong solution:** X_t adapted to the filtration generated by the driving Brownian motion W_t .
- **Weak solution:** existence of pair (X, W) on some probability space satisfying the SDE (distributional notion).
- **Existence & uniqueness:** If μ and σ satisfy global Lipschitz and linear growth conditions, then the SDE has a unique strong solution.
- **Generator (infinitesimal):** for smooth f , $\mathcal{L}f(x) = \mu(t, x)f'(x) + \frac{1}{2}\sigma^2(t, x)f''(x)$.

- **Fokker–Planck (forward) PDE:** density $p(t, x)$ solves $\partial_t p = -\partial_x(\mu p) + \frac{1}{2}\partial_{xx}(\sigma^2 p)$.

Ito Integral

- **Ito integral:** For Brownian motion $(W_t)_{t \geq 0}$, the stochastic integral is

$$I_g(t) = \int_0^t g(s) dW_s$$

It is defined as the L^2 -limit of left-point sums $\int_0^t g(s) dW_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(t_i)(W_{t_{i+1}} - W_{t_i}), \quad t_i = \frac{it}{n}.$

Important: Brownian motion is not differentiable, so generally

$$\int_0^t g(s) dW_s \neq \int_0^t g(s) W'(s) ds.$$

- **Mean zero and Ito isometry:** If g is adapted and square-integrable, then: $E\left[\int_0^t g(s) dW_s\right] = 0$, and

$$E\left[\left(\int_0^t g(s) dW_s\right)^2\right] = E\left[\int_0^t g(s)^2 ds\right]$$

- **Useful Ito rules:** For (standard) Brownian motion W_t , $W_0 = 0$, $(dW_t)^2 = dt$, $dt dW_t = 0$, $(dt)^2 = 0$.

These rules are the reason why Ito calculus differs from ordinary calculus.

Ito's Formula

- **Ito's formula:** If $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ and $Y_t = f(t, X_t)$ with $f \in C^{1,2}$, then $dY_t = \left(f_t + \mu f_x + \frac{1}{2}\sigma^2 f_{xx}\right) dt + \sigma f_x dW_t$.

- **Special case:** $X_t = W_t \rightarrow$ Brownian Motion, $\mu = 0, \sigma = 1$ $dY_t = \left(f_t(t, W_t) + \frac{1}{2}f_{ww}(t, W_t)\right) dt + f_w(t, W_t)dW_t$.

- **Martingale trick (PDE):** If the drift term is zero, then the process is a martingale candidate. For Brownian motion, if $f_t(t, w) + \frac{1}{2}f_{ww}(t, w) = 0$,

then $f(t, W_t)$ is a martingale, assuming integrability.

Ito Calculation Rules

Be careful: it is not only important whether the differential is ds , dW_s , or dX_s . It is also important what kind of expression is being integrated.

- **Time integrals:** Integrals with respect to ds are ordinary time integrals. For example, $\int_0^t 2s ds = [s^2]_0^t = t^2$.
- **Ito integrals:** Integrals with respect to dW_s are Ito integrals and cannot be treated like ordinary integrals: $\int_0^t W_s dW_s \neq \frac{1}{2}W_t^2$. So if the integrand contains W_s , usually leave it as an integral: $\int_0^t W_s ds$.

- **Process differentials:** If the integrand is a full differential, then integrate directly:

$$\int_0^t dX_s = X_t - X_0.$$

For example, since $W_0 = 0$,

$$\int_0^t d(W_s^2) = W_t^2 - W_0^2 = W_t^2.$$

- **Brownian moments and Fubini:** For $W_s \sim \mathcal{N}(0, s)$, $E\left[\int_0^t W_s^n ds\right] = \int_0^t E[W_s^n] ds$, $E[W_s^{2k+1}] = 0$, $E[W_s^{2k}] = (2k - 1)!! s^k$. In particular, $E[W_s^2] = s$, $E[W_s^4] = 3s^2$, $E[W_s^6] = 15s^3$.
- **Quadratic variation:** For Brownian motion, $[W]_t = t$. If $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, then $[X]_t = \int_0^t \sigma(s, X_s)^2 ds$.

Problem-Solving Considerations

- Identify SDE class: linear, multiplicative (GBM), additive noise, OU, CIR, or general.
- Check Lipschitz/Growth conditions for existence & uniqueness (Picard iterations).
- Use Itô's lemma for transformations: pick $f(t, x)$ so that resulting SDE is integrable.
- For linear SDEs, first classify the noise: additive noise $\rightarrow$ deterministic integrating factor; multiplicative noise $\rightarrow$ log-transform or stochastic integrating factor.
- For multiplicative GBM with constant α, β ,

$$\begin{aligned} dX_t &= \alpha X_t dt + \beta X_t dW_t: \\ &\Rightarrow X_t = X_0 \exp\left(\left(\alpha - \frac{1}{2}\beta^2\right)t + \beta W_t\right). \end{aligned}$$

- **Time-dependent multiplicative SDE:** For $dX_t = \alpha(t)X_t dt + \beta(t)X_t dW_t$, $X_t = X_0 \exp\left(\int_0^t \left(\alpha(s) - \frac{1}{2}\beta(s)^2\right) ds + \int_0^t \beta(s) dW_s\right)$.

- Use moment equations: apply Itô to $f(x) = x^k$ and take expectations (Itô integral mean zero).
- To get stationary densities use Fokker–Planck steady state when time-homogeneous.

Linear SDE types: A linear SDE has drift and diffusion terms that are at most linear in X_t . For the additive-noise form $dX_t = a(t)X_t dt + b(t) dt + \sigma(t) dW_t$, the dW_t -coefficient does not depend on X_t , so a deterministic integrating factor works. If the noise is multiplicative, e.g. $dX_t = a(t)X_t dt + c(t)X_t dW_t$, then the dW_t -coefficient also contains X_t ; use a log-transform for GBM or a stochastic integrating factor.

Solving linear SDEs with integrating factor: For a linear SDE $dX_t = a(t)X_t dt + b(t) dt + \sigma(t) dW_t$, choose

$$Y_t = \exp\left(-\int_0^t a(u) du\right).$$

Use Ito's special case formula to find $dY_t = -a(t)Y_t dt$. Then Use Ito's product rule: $d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$.

Here dY_t has no dW_t -part, so $dX_t dY_t = 0$. The $a(t)X_t Y_t dt$ -terms cancel, giving $d(X_t Y_t) = b(t)Y_t dt + \sigma(t)Y_t dW_t$.

Integrate from 0 to t :

$$X_t Y_t - X_0 Y_0 = \int_0^t b(s) Y_s ds + \int_0^t \sigma(s) Y_s dW_s.$$

Since $Y_0 = e^0 = 1$, solve for X_t : $X_t = e^{\int_0^t a(u) du} \times \left(X_0 + \int_0^t e^{-\int_0^s a(u) du} b(s) ds + \int_0^t e^{-\int_0^s a(u) du} \sigma(s) dW_s\right)$

This method works for linear SDEs when a suitable integrating factor can be found.

Geometric Brownian Motion (GBM) / log-transform: For $dX_t = \alpha X_t dt + \beta X_t dW_t$ with constant α, β , use $Y_t = \log X_t$. By Ito's formula,

$$d \log X_t = \left(\alpha - \frac{1}{2}\beta^2\right) dt + \beta dW_t.$$

Hence

$$X_t = X_0 \exp\left(\left(\alpha - \frac{1}{2}\beta^2\right)t + \beta W_t\right).$$

Examples & How to Solve

- **Ornstein–Uhlenbeck (OU):** $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$. Solution: $X_t = X_0 e^{-\theta t} + \mu(1 - e^{-\theta t}) + \sigma \int_0^t e^{-\theta(t-s)} dW_s$.

- **Cox–Ingersoll–Ross (CIR):** $dX_t = \kappa(\theta - X_t)dt + \sigma\sqrt{X_t}dW_t$ (nonlinear; positivity-preserving under Feller condition).
- **Linear SDE with multiplicative noise:** solve using exponential integrating factor; include Itô correction $-\frac{1}{2}$ term when transforming logs.
- **Itô integral evaluation:** pick f so that $f'(x)$ matches integrand and apply Itô to express integral in terms of $f(\cdot)$ and time integral.
- **Moment dynamics:** for mean $m(t) = E[X_t]$, compute using $dE[X_t] = E[\mu(t, X_t)]dt$; for variance use Itô on X_t^2 and expectations.

Probability generating functions and branching processes

- **Probability generating function (PGF):** For $X \in \mathbb{N}_0$, $G_X(s) = E[s^X] = \sum_{i=0}^\infty s^i P(X = i)$.

Moments can be obtained from derivatives: $E[X] = G_X'(1)$ and $\text{Var}(X) = G_X''(1) + G_X'(1) - (G_X'(1))^2$.

- **Branching process:** Let Z_n be the population size in generation n , with $Z_0 = 1$. If G is the PGF of the offspring distribution, then the PGF G_n of Z_n satisfies $G_n(s) = G_{n-1}(G(s))$.

- **Extinction probability:** The ultimate extinction probability η is the smallest solution in $[0, 1]$ of $s = G(s)$.
- If $m = G'(1) \leq 1$ (subcritical or critical), then $\eta = 1$. If $m > 1$ (supercritical), then usually $\eta < 1$.
- **Valid PGF check:** A function $G(s) = \sum_{k=0}^\infty p_k s^k$ is a valid probability generating function if $p_k \in [0, 1]$ for all k and $\sum_{k=0}^\infty p_k = 1$. **Equivalently**, $G(1) = 1$ and $G(s) \in [0, 1]$ for $s \in [0, 1]$.
- **Extinction by generation n :** Let $\eta_n = P(Z_n = 0)$. If $Z_0 = 1$, then $\eta_0 = P(Z_0 = 0) = 0$, $\eta_n = G(\eta_{n-1})$.

Thus $\eta_1 = G(0) = P(\text{offspring} = 0)$, and $P(\text{extinction exactly at generation } n) = \eta_n - \eta_{n-1}$.